

Files

1 Basic Idea

Definition

A **file** is an operating-system abstraction for storing information on secondary storage and reading it back later.

- A file hides the physical details of disks and storage devices.
- Users do not need to know where the data are physically stored.
- Users do not need to know how the disk hardware works.
- A process creates a file and gives it a name.
- The file can continue to exist after the process that created it terminates.
- Other processes can later access the file by using its name.

Core Point

The file abstraction separates the user's view of stored information from the hardware details of storage.

2 File Naming

Why Naming Matters

Naming is one of the most important parts of any abstraction because names are how users and processes refer to stored objects.

- File-naming rules vary between operating systems.
- Most operating systems allow alphabetic strings as file names.
- Examples: `andrea`, `bruce`, `cathy`.
- Many systems also allow digits and special characters.
- Examples: `2`, `urgent!`, `Fig.2-14`.
- Many modern file systems support file names up to about 255 characters.

3 Case Sensitivity

Case Sensitivity

A file system is **case-sensitive** if uppercase and lowercase letters are treated as different characters in file names.

System type	Meaning	Example
Case-sensitive	Uppercase and lowercase names can refer to different files.	UNIX can treat <code>maria</code> , <code>Maria</code> , and <code>MARIA</code> as three distinct files.
Case-insensitive	Uppercase and lowercase versions of a name refer to the same file.	MS-DOS treats <code>maria</code> , <code>Maria</code> , and <code>MARIA</code> as the same file.

Common Mistake

Do not assume that file names behave the same way on every operating system. Case sensitivity depends on the file system and OS rules.

4 Windows File-System Note

- Older MS-DOS used the FAT file system.
- Windows 95 and Windows 98 used FAT-16 and later FAT-32 extensions.
- Newer Windows versions support FAT variants for compatibility.
- Newer Windows systems also use NTFS as a more advanced native file system.
- NTFS supports features not found in old FAT systems, such as richer metadata and Unicode file names.
- exFAT is a FAT-like Microsoft file system designed for flash drives and large file systems.

5 File Extensions

Definition

A **file extension** is the part of a file name after a period, often used to indicate the file's type or intended use.

- Many systems support two-part file names such as `prog.c`.
- The part before the period is the main name.
- The part after the period is the extension.
- MS-DOS used names of 1 to 8 characters plus an optional 1 to 3 character extension.
- UNIX allows extensions by convention and does not restrict them to three characters.
- UNIX file names may contain multiple extensions, such as `homepage.html.zip`.

6 Common Extensions

Extension	Meaning
<code>.bak</code>	Backup file.
<code>.c</code>	C source program.
<code>.gif</code>	GIF image file.
<code>.hlp</code>	Help file.
<code>.html</code>	HTML Web document.
<code>.jpg</code>	JPEG still picture.
<code>.mp3</code>	MPEG layer 3 audio file.
<code>.mpg</code>	MPEG movie file.
<code>.o</code>	Object file produced by a compiler, not yet linked.
<code>.pdf</code>	Portable Document Format file.
<code>.ps</code>	PostScript file.
<code>.tex</code>	Input file for the $\text{\TeX}$ formatting system.
<code>.txt</code>	General text file.
<code>.zip</code>	Compressed archive.

7 Extensions as Conventions

- In UNIX-like systems, extensions are usually conventions.
- The operating system generally does not enforce the meaning of an extension.
- A file named `file.txt` is expected to be text, but the OS does not guarantee it.
- Programs may still enforce their own extension rules.
- A C compiler may require source files to end in `.c`.
- Extensions help programs distinguish different input file types.

Compiler Example

A compiler can use extensions to distinguish C source files, assembly-language files, object files, and other input files.

8 Extensions with OS Meaning

- Windows assigns more direct meaning to file extensions.
- Users or programs can register an extension with the operating system.
- The registration says which application owns that extension.
- When the user double-clicks a file, Windows can launch the associated application.
- Example: double-clicking `file.docx` opens Microsoft Word with that file.

9 UNIX vs. Windows View

Feature	UNIX-like systems	Windows-style systems
Case handling	Often case-sensitive.	Often case-insensitive for common user file systems.
Extension meaning	Mostly a convention for users and programs.	Often registered with the OS and tied to applications.
Multiple extensions	Common and allowed, such as <code>file.html.zip</code> .	Allowed, but the last extension often drives association.
OS enforcement	OS usually does not care whether content matches extension.	OS often uses the extension to choose the default application.

Final Point

Files give users a named, persistent storage abstraction. File-name rules, case sensitivity, and extension meaning depend on the operating system and file system.